GuardianNav: ML-Driven Safe and Personalized Navigation for Women.

EPICS Project report submitted in partial fulfillment of the Requirements for the Award of the Degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

by

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VELAGAPUDI RAMAKRISHNA SIDDHARTHA ENGINEERING COLLEGE

(Autonomous, Accredited with 'A+' grade by NAAC)

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CERTIFICATE

This is to certify that the EPICS Project Report entitled **"GuardianNav: ML-Driven Safe and personalized Navigation for Women"** being submitted by

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in partial fulfilment for the award of the Degree of Bachelor of Technology in Computer Science and Engineering to the Jawaharlal Nehru Technological University, Kakinada, is a record of bonafide work carried out during the period from 2024 - 2025.

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DECLARATION

We hereby declare that the EPICS Project entitled **"GuardianNav: ML-Driven Safe and Personalized Navigation for Women"** submitted for the B.Tech Degree is our original work and the dissertation has not formed the basis for the award of any degree, associateship, fellowship or any other similar titles.

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Table of Contents

List of Figures

Abstract

In today's world, personal safety during travel is a critical concern, especially for women who often face heightened risks in unfamiliar or vulnerable environments. To address this challenge, GuardianNav is introduced as a comprehensive navigation app designed to prioritize safety and empower users with confidence in their journeys. GuardianNav provides personalized and secure route recommendations by analysing diverse data sources, including crime data sources, userreported incidents, and real-time safety updates. The app integrates key features such as Safety Scores for route evaluation, Emergency Alerts to notify contacts in distress, Historical Safety Trends for insights into safety changes in frequently travelled areas. Furthermore, the app delivers Personalized Recommendations, adapting routes based on user preferences and dynamic conditions. By leveraging community-driven feedback, user insights, and machine learning techniques, GuardianNav ensures continuous enhancement to meet user needs. Through this innovative approach, GuardianNav aims to redefine travel safety, particularly for women, by offering an informed, secure, and empowering navigation experience.

Keywords: GuardianNav, personalized, safe route recommendations, navigation app, Machine Learning, K-Means Clustering, Crime, Crime Data

Chapter 1

INTRODUCTION

1.1 Introduction

In today's world, personal safety during travel is a growing concern, particularly for women who face unique challenges and heightened risks in various environments. With the rapid expansion of urban areas and the complexities of modern life, it has become increasingly critical to develop solutions that prioritize security and empower individuals to travel confidently. Addressing these challenges requires innovative approaches that combine technology, data analysis, and community involvement. This project introduces GuardianNav, a cutting-edge navigation application specifically designed to enhance travel safety. Leveraging advanced machine learning technologies, GuardianNav provides users with secure and personalized route recommendations. By analyzing crime statistics, user-reported incidents, and real-time safety data, the app aims to minimize risks and promote safer travel experiences.

- **Safety Scores for Routes:** Routes are scored based on crime patterns, time of day, and incident trends, giving users a clear understanding of the safest options.
- **Emergency Contact Alerts:** Immediate notifications to pre-registered contacts in case of emergencies.
- **Safe Haven Identification:** Identifying nearby locations such as police stations, hospitals, or certified safe spaces to seek help if needed
- **Real-Time Incident Reporting and Alerts:** Routes are scored based on crime patterns, time of day, and incident trends, giving users a clear understanding of the safest options.
- **Personalized Route Recommendations:** Enabling users to report and receive updates on safety-related incidents in their vicinity.
- **Historical Safety Trends:** Tailored suggestions based on individual preferences, historical data, and current safety conditions.

A distinctive aspect of GuardianNav is the application of K-means clustering to optimize route selection. This clustering algorithm identifies groups of safe routes and ensures that the shortest path among them is highlighted, balancing efficiency and security. The app also fosters community engagement by incorporating user feedback and insights, creating a dynamic and evolving safety ecosystem. By integrating diverse data sources, such as local law enforcement records, social media reports, and community alerts, GuardianNav builds a comprehensive safety framework. In addition to addressing women's travel safety, the app can be adapted to serve broader audiences, including senior citizens, children, and individuals in unfamiliar areas. Its versatility makes it a valuable tool for enhancing safety awareness and navigation worldwide. GuardianNav represents a significant step forward in addressing modern travel challenges. By combining machine learning, real-time data, and user-centric features, this app seeks to empower individuals, particularly women, to navigate their journeys with confidence, transforming personal travel into a safer and more informed experience.

1.2 Motivation

Women navigating urban environments often face heightened safety concerns, yet existing navigation apps prioritize distance and traffic over security. This lack of safetyfocused features leaves users vulnerable to risks such as harassment or assault, especially in potentially unsafe areas. Without access to real-time crime data, user-reported incidents, and dynamic safety assessments, women are left without adequate information to make informed travel decisions. The gap in safety-centric navigation tools exacerbates these risks, limiting confidence and security during travel. There is a critical need for a solution that incorporates comprehensive safety data to provide personalized, safer route recommendations. Addressing this issue is essential to empowering women and promoting safer travel experiences.The GuardianNav can be very helpful for people.

1.3 Visit to Client

Figure 1.1: Client Location

Figure 1.1 shows route to Client Location-Raj Bhavan, Governorpet, Vijayawada-520002 from V.R Siddhartha Engineering College.

1.4 Photo with Client

Figure 1.2: Meeting with Client

Figure 1.2 shows the images with our Client Mr. Y.V.V.L Naidu who is currently serving as Circle Inspector at Nandigama Division.

1.5 Problem Statement

In urban environments, women often face heightened concerns about personal safety while navigating through public spaces. Existing navigation apps primarily focus on optimizing routes based on factors like distance and traffic but overlook safety-related considerations. There is a critical need for a navigation solution that addresses the specific safety concerns of women, incorporating real-time crime data, user reports, and dynamic safety assessments to provide safer route recommendations. Without such a solution, women are left vulnerable to navigating through potentially unsafe areas without adequate information, increasing the risk of harassment, assault, or other forms of danger.

1.6 Relevance

GuardianNav is highly relevant in today's world, where urban safety has become a pressing concern, especially for women and vulnerable groups. Traditional navigation systems prioritize efficiency over security, leaving users unaware of potential risks along their routes. By integrating real-time crime data from sources like the UK Police API, GuardianNav fills this critical gap, offering safety-first navigation that identifies and avoids high-risk zones. Its ability to dynamically adapt routes based on updated crime data ensures that users have access to the most secure travel options at all times. With its focus on personalized safety, GuardianNav provides an essential tool for navigating urban environments, empowering users to make informed decisions and enhancing confidence during travel. Its relevance extends beyond individual users, contributing to broader public safety initiatives and fostering safer communities.

1.7 Scope

The proposed solution aims to address the critical need for enhanced personal safety for women navigating urban spaces by developing a safety-focused navigation application. The scope of this solution includes the following aspects:

- Utilizing UK Police API data to map crime trends in London effectively.
- Offering an intuitive user interface to display safe routes and real-time alerts.
- Empowering users to navigate with confidence by avoiding crime hotspots.

This innovation not only addresses the immediate need for safer navigation but also sets the groundwork for scalable applications across different regions, revolutionizing travel safety for vulnerable groups and urban commuters alike.

1.8 Objectives

The Objectives of our project are:

1. Ensure safer travel by avoiding high-risk zones identified using crime data and clustering algorithms.
2. Analyze and process crime data using clustering algorithms (e.g., KMeans) to identify high-risk areas and generate actionable insights for safe route recommendations.
3. Develop an intuitive and interactive user interface that enables users to input locations, view safe routes, and receive real-time alerts, ensuring a seamless and user-friendly experience.

Chapter 2

LITERATURE REVIEW

2.1 Algorithmic Crime Prediction Model Based on the Analysis of Crime Clusters.

A. Malathi, Dr. S. Santhosh Baboo, D.G. Vaishnav College, Chennai[1]. Crime is a complex behavioral issue influenced by a combination of social, economic, and environmental factors. It poses significant challenges to society, causing both tangible and intangible losses. Addressing crimes effectively and developing methods to solve them more efficiently can lead to substantial societal benefits. This paper explores the application of data mining techniques to analyze crime data, specifically focusing on handling missing values and clustering patterns.

The study examines the use of the Missing Value (MV) Algorithm and an enhanced version of the Apriori Algorithm to address two critical challenges: filling in incomplete data entries and identifying meaningful crime patterns. The MV Algorithm is applied to impute missing data within the crime records, ensuring the dataset is complete and reliable for analysis. Meanwhile, the Apriori Algorithm, typically used for association rule mining, is enhanced to detect correlations between crime attributes, aiding in the discovery of crime trends and hotspots.

Additionally, a semi-supervised learning technique is employed to extract insights from crime records, leveraging both labeled and unlabeled data. This approach improves knowledge discovery and enhances the predictive accuracy of crime analysis. This research underscores the importance of combining data preprocessing techniques, such as missing value handling, with advanced algorithms like Apriori and semi-supervised learning to gain actionable insights from crime data, ultimately contributing to more effective crime-solving strategies.

2.2 ML-Driven Navigation Systems for Personalized Travel Safety

Ali, A., Joseph, A. [2] developed a system where machine learning algorithms analyze user behavior and historical safety data to suggest personalized travel routes.

Advantages:

- Offers highly personalized safety recommendations based on user preferences.
- Machine learning allows continuous system improvement through user feedback.

Disadvantages:

- ML models may require large datasets to be fully effective, which could impact accuracy for new or sparse data regions.

This approach highlights the potential of machine learning to revolutionize navigation systems, offering dynamic and personalized solutions and underlining challenges related to data quality and computational requirements.

2.3 Clustering Algorithms for Route Optimization in Safety Navigation Applications

Gupta, R., Patel, S. [3] explored the application of clustering algorithms in route optimization to enhance travel safety. Their approach integrates real-time traffic data with safety metrics to identify high-risk zones and provide safer alternative routes for users. The system dynamically adjusts based on the latest data, ensuring optimized paths that balance safety and efficiency.

Advantages:

- Identification of Unsafe Zones: Clustering algorithms effectively detect and categorize high-risk areas, helping users avoid these zones and reduce potential travel risks.
- Dynamic Route Optimization: Routes are recalculated in real time, considering factors like safety and time, ensuring a balance between efficiency and security.

Disadvantages:

- Data Dependency: The effectiveness of clustering algorithms diminishes if they are not recalibrated frequently with updated data, potentially leading to outdated or inaccurate risk assessments.
- Computational Challenges: Handling large-scale, real-time data may pose computational challenges, particularly in densely populated or high-traffic regions.

This study underscores the utility of clustering techniques in identifying unsafe areas and optimizing routes for safer navigation, while also highlighting the need for regular updates and robust computational frameworks to maintain accuracy and reliability.

2.4 Enhancing Women's Safety Through Navigation Systems Using Real-Time Data and Alerts

Anthony, M., & Varia, R. [4] explored the role of real-time data and alert systems in improving women's safety during navigation. Their research emphasizes the use of crowdsourced safety information and community-driven feedback to identify potential risks, allowing the navigation system to make immediate route adjustments in response to threats or incidents.

Advantages:

- **Crowdsourced Data Utilization:** Real-time user-generated data enhances the system's ability to detect unsafe conditions in urban settings, providing updated and accurate safety insights.
- **Instant Alert Mechanisms:** The inclusion of alerts enables users to quickly respond to potential threats, ensuring proactive safety measures.

Disadvantages:

- **Dependence on User Participation:** The accuracy and reliability of the system heavily depend on active user engagement. In areas with low participation or sparse data, its effectiveness may be compromised.
- **Data Overload Risks:** Managing and filtering large volumes of real-time data from multiple users can present challenges, potentially delaying response times.

This research highlights the transformative potential of integrating real-time data and alerts into navigation systems to enhance women's safety. However, it also underscores the need for widespread user participation and robust data handling mechanisms to ensure system reliability and effectiveness.

2.5 Implementation of Real-Time Safety Scoring in Navigation Systems

Bose, S., Dasgupta, A., & Chakraborty, R. [5] proposed a navigation system that integrates a real-time safety scoring mechanism to enhance route selection for users. The system calculates safety scores for various routes based on live incident data, historical safety metrics, and environmental factors. These scores dynamically update as new safety reports are received, ensuring the system reflects the most current safety conditions. By offering transparent safety ratings for all available routes, users can make informed decisions tailored to their security preferences.

Advantages:

- **Dynamic Safety Assessment:** The system constantly monitors and recalculates safety scores based on incoming data, maintaining relevance and accuracy in its recommendations.
- **Transparent Decision-Making:** Users are provided with clear safety ratings, which build trust and empower them to prioritize routes aligning with their safety needs.
- **Enhanced Adaptability:** Real-time scoring enables the system to respond quickly to changes in safety conditions, such as sudden incidents or updates in high-risk zones.

Disadvantages:

- **High Data Dependency:** The system's effectiveness depends on consistent and frequent updates from reliable real-time data sources. Limited or delayed data can compromise accuracy.
- **Resource-Intensive:** Processing and maintaining large volumes of real-time data require robust computational infrastructure, increasing operational complexity and cost.
- **Scalability Challenges:** The need for constant updates and scoring adjustments may pose difficulties in areas with limited technological access or infrastructure.

This system demonstrates the transformative potential of real-time safety scoring in navigation by fostering safer route selection and adapting dynamically to live data. However, it also highlights critical challenges related to data availability, resource management, and system scalability that must be addressed to ensure long-term efficacy.

2.6 Real-Time Navigation Enhancements with Machine Learning

Smith, J., & Williams, A. [6] investigated the application of machine learning techniques to improve real-time navigation systems. Their approach integrates multiple data streams, including traffic conditions, safety metrics, and environmental factors, to generate optimized and adaptive route recommendations. By leveraging machine learning, the system continuously refines its predictions and adapts to dynamic travel conditions in real-time.

Advantages:

- **Continuous Improvement:** Machine learning models learn from historical and real-time data, enabling ongoing refinement of navigation predictions and increasing accuracy over time.
- **Adaptive Route Planning:** The system dynamically adjusts routes based on real-time inputs, ensuring optimal travel efficiency and enhanced safety for users.
- **Holistic Integration:** By combining diverse data types—such as traffic congestion, accident reports, and weather conditions—the system delivers more contextually aware recommendations.

Disadvantages:

- **Data Dependency:** The system's performance is highly reliant on external data sources, which may be inconsistent or unavailable in certain regions.
- **Quality and Latency Issues:** Poor-quality data or delays in data updates can hinder the accuracy and responsiveness of the navigation system.
- **Resource Intensive:** Real-time processing of vast datasets requires advanced computational infrastructure, which may not be feasible in all scenarios.

This research highlights the potential of machine learning to revolutionize real-time navigation by improving route efficiency and user safety. However, it also underscores the importance of robust data pipelines and computational resources to mitigate challenges related to data dependency and system scalability.

2.7 Safe Route Recommendations with Clustering Algorithms: A Comparative Analysis

Liu, Y., & Zhang, T. [7] conducted an in-depth study to evaluate the effectiveness of various clustering algorithms in generating safe route recommendations for urban travelers. Their research focuses on identifying which clustering techniques are most suitable for segmenting urban areas based on safety metrics derived from crime data, traffic incidents, and other real-time factors. By performing a comparative analysis, the study aims to highlight the strengths and limitations of different clustering methods in addressing the complexities of safety-based navigation.

Advantages:

- **Algorithm Comparison:** The study systematically compares multiple clustering algorithms, such as K-Means, DBSCAN, and Hierarchical Clustering, to identify the most effective method for segregating high-risk zones and suggesting safer routes.

- **Enhanced Safety Recommendations:** The insights from this comparative analysis help refine clustering techniques, leading to more accurate identification of unsafe areas and improved route suggestions for users.
- **Improvement in Clustering Techniques:** The research offers practical guidance for optimizing clustering algorithms, making them more adaptable to real-world urban navigation challenges.

Disadvantages:

- **Urban Complexity:** Urban areas are characterized by dynamic and diverse conditions, such as fluctuating population densities and constantly changing traffic patterns. Some clustering algorithms may struggle to handle this variability effectively.
- **Optimization Requirements:** The algorithms often require fine-tuning of parameters and frequent recalibration to ensure accuracy, especially when applied to complex datasets with mixed safety indicators.
- **Computational Overheads:** Advanced clustering techniques, particularly those analyzing large datasets in real-time, can be resource-intensive, posing scalability issues for widespread implementation.

Key Findings and Applications: The study reveals that:

- K-Means Clustering performs well in generalizing safety zones but may overlook irregularly shaped clusters, such as crime hotspots along specific roadways.
- DBSCAN is effective in identifying dense clusters of unsafe areas but can be sensitive to parameter settings, impacting its adaptability to diverse environments.
- Hierarchical Clustering provides detailed insights into nested patterns of safety risks but is computationally expensive and less practical for real-time applications.

The findings suggest that a hybrid approach, combining the strengths of different algorithms, could be most effective for urban safety navigation. For instance, K-Means could be used for initial clustering, while DBSCAN refines the boundaries of high-risk zones. Liu and Zhang's analysis highlights the importance of algorithmic selection and optimization in ensuring accurate and actionable safety recommendations. Their work contributes to the advancement of clustering-based navigation systems, helping to create safer travel experiences in complex urban environments.

2.8 Key Takeaways from the Literature Survey

- **Data-Driven Safety Enhancements:**
  - Leveraging data mining, clustering, and machine learning techniques provides actionable insights for navigation and safety applications.
  - Real-time data integration enables dynamic route adjustments, improving both safety and travel efficiency.
- **Personalized Navigation Systems:**
  - Personalization through user behavior analysis and historical safety data enhances relevance and user trust.
  - Machine learning and clustering algorithms continuously refine recommendations, adapting to changing conditions.
- **Clustering for Risk Identification:**
  - Clustering techniques are effective for identifying high-risk zones and optimizing safer routes.
  - Combining different algorithms or using hybrid approaches can address urban complexities and improve accuracy.
- **Real-Time Scoring and Alerts:**
  - Safety scoring mechanisms provide transparent, user-friendly safety ratings, empowering informed decision-making.
  - Immediate alerts based on live data ensure rapid responses to potential safety threats.
- **Crowdsourced and Community-Driven Data:**
  - Crowdsourced data enhances safety recommendations but depends on active user participation.
  - Community-driven feedback can enrich safety assessments but requires robust mechanisms for filtering and validation.
- **Challenges in Data Dependency:**
  - All systems rely on high-quality, consistent, and frequently updated data.
  - Sparse or outdated data reduces the effectiveness of predictive models and safety assessments.

- **Computational and Scalability Concerns:**
  - Real-time processing and handling large datasets demand significant computational resources.
  - Scalability remains a challenge, particularly in densely populated or under-resourced regions.
- **Continuous Improvement:**
  - Machine learning systems and clustering algorithms must evolve with user feedback and new data inputs.
  - Regular recalibration and parameter tuning are essential to maintain system accuracy and reliability.
- **User-Centric Design:**
  - Navigation systems must prioritize user needs, balancing safety with convenience and efficiency.
  - Transparent communication of safety metrics builds trust and engagement.
- **Future Opportunities:**
  - Combining multiple algorithms and data sources can create more robust and adaptive safety systems.
  - Addressing challenges in data quality, computational efficiency, and user participation will be critical for advancing navigation technologies.

Chapter 3

SOFTWARE REQUIREMENTS ANALYSIS

This chapter describes the functional and non-functional requirements of the project.

3.1 Functional Requirements

Functional Requirements specify system behavior regarding what the system does for any action. Functional requirements define product features and focus on user requirements. The requirements include use cases, user stories, prototypes, etc. They determine if the system requires any computations, inputs, and outputs.

3.1.1 Software Requirements

The list of software requirements for the project includes the following:

- **Programming Languages:** Python, JavaScript
- **Environments:** Flask Framework, Leaflet.js, OpenRouteService API
- **Libraries used:** folium, geopy, openrouteservice, pandas, scikit-learn

3.2 Non-Functional Requirements

Non-functional requirements define system behavior, features, and general characteristics that affect the user experience. How well non-functional requirements are defined and executed determines how easy the system is to use, and is used to judge system performance. Non-functional requirements are product properties and focus on user expectations.

3.2.1 Performance

1. **Map Rendering and Routing:**
   - The Leaflet.js library efficiently renders interactive maps with smooth zooming, panning, and overlay capabilities.
   - Leaflet Routing Machine ensures fast route generation and visualization using OpenRouteService API, allowing users to dynamically set start and end points.

- Real-time marker updates and route recalculations happen seamlessly, provided the internet connection and API response are fast.

2. **Real-Time Data Handling:**

   - **Crime Data Visualization:**
     - High-risk areas and crime locations are plotted using clustering algorithms (K-Means) and geospatial data.
     - Markers and popups on the map provide immediate context, enabling users to make informed decisions about safety.
   - The dynamic addition of new markers for high-risk areas ensures the map remains up-to-date and informative.

3. **Backend Processing:**

   - Route Safety Optimization: The backend uses K-Means clustering to identify high-risk clusters and filters unsafe routes effectively.
   - The Flask-based server handles user requests and API responses efficiently for small to medium workloads. However, scalability may require load balancing or cloud deployment for higher traffic.
   - The Python libraries (geopy, openrouteservice, etc.) provide robust calculations for route safety, although computation time increases with larger datasets.

4. **Responsiveness:**

   - The application dynamically adjusts the map and route based on user input, ensuring a responsive user experience.
   - Real-time geocoding (via Nominatim API) and route fetching are subject to external API latencies but are generally quick for single user operations.

5. **Scalability:**

   - The application is designed to scale by incorporating larger datasets and adapting to high user demand.
   - The use of clustering algorithms like K-Means ensures that even large datasets are processed efficiently, though adding more clusters may increase computation time.

6. **Limitations:**

   - **Real-Time Data Dependency:** The application's performance is highly reliant on the availability and responsiveness of external APIs (e.g., Nominatim, OpenRouteService).

- **Cluster Tolerance:** Strict avoidance of high-risk clusters may occasionally result in limited or no route suggestions.
- **System Load:** For very large datasets or high traffic, the Flask server and clustering algorithm may experience delays without optimization or distributed computing.

7. **Optimization Opportunities:**

   - Caching frequently accessed geocoding results and API responses can reduce response time.
   - Migrating from Flask to a more scalable framework (e.g., FastAPI) or hosting on cloud platforms with autoscaling can improve performance under high traffic.

Overall, the GuardianNav code is designed for reliable and efficient performance in small to medium-scale deployments, with clear opportunities for optimization and scaling to support broader usage.

Chapter 4

Proposed System

4.1 Problem Understanding

A 2021 UN Women report stated that almost 58% of the women population globally feel unsafe navigating alone at night in their respective neighborhoods. Fear of theft, harassment, or violence often influences women's decisions about routes and travel times. Current navigation tools like Google Maps focus on the shortest routes, ignoring safety metrics like crime-prone zones or well-lit streets. Many existing platforms lack real-time updates on safety risks, leaving users uninformed about potential dangers.

The features we typically need in addition to navigation, but do not receive, are as follows:

- Absence of Safety Metrics.
- Lack of Co-ordinate Crime Data Integration.
- Inadequate Community Reporting.
- No utilization of local law enforcement data.

The impacts that the public might encounter due to the above inabilities are as follows:

- **Reduced Mobility:** Women may limit travel after dark or avoid areas perceived as unsafe, restricting their freedom of movement. This reduces their access to opportunities like late-night shifts, networking events, or education programs.
- **Increased Exposure to Risk:** Lack of safety-focused navigation may inadvertently direct users through unsafe areas, increasing their vulnerability to crime.
- **Economic and Professional Impact:** Unsafe commutes can cause businesses and organizations to lose potential talent and workforce.
- **Psychological Stress:** Constant vigilance and fear of navigating unknown or unsafe areas lead to anxiety, stress, and reduced mental well-being.

4.2 Ideation and Concept Development

Therefore, GuardianNav's main concept is to enhance navigation systems by integrating machine learning to recommend safer routes based on personalized information and real-world concerns. The goal is to promote safe and happy navigation and empower users, particularly women, with peace of mind during travel.

Navigation integrated with safety can promote productivity and encourage:

- **Increased Mobility and Confidence:** Individuals, particularly women, feel more confident exploring new areas without fear and harm. Opportunities are also increased where the routes weren't used or visited due to safety concerns.
- **Better Resource Utilization:** Reliable safety data reduces delays caused by rerouting through unsafe areas, optimizing travel time. Roads are more utilized and more connectivity is ensured through areas.
- **Improved Urban Planning and Safety Awareness:** Data-driven decisions can be facilitated to guide policymakers and urban planners to improve infrastructure and lighting in high-risk zones. Collective responsibility can also be expected through engagement with users' feedback.

Unique Selling Point of GuardianNav: The Unique Selling Point (USP) of GuardianNav lies in its ability to provide safe, personalized navigation by integrating real-time crime data and user-generated safety reports into the routing system. Unlike traditional navigation apps, GuardianNav evaluates and recommends routes based on safety metrics, such as nearby crime incidents and reported hazards in unsafe areas. The app's personalized safety preferences allow users to customize their routes to their unique safety concerns, such as avoiding poorly lit streets or high-crime neighborhoods. With an intuitive interface, GuardianNav ensures that users receive safer travel options, making it a standout solution for women seeking security and efficiency in their daily commutes.

4.3 Design Thinking Process

This process ensures that GuardianNav's app addresses real concerns surrounding safe navigation for women, combining empathy, creativity, and practicality. By following the key stages—Empathize, Define, Ideate, Prototype, and Test—the development of GuardianNav is rooted in user feedback, real-world data, and iterative improvements. This approach enhances the app's usability and ensures that its features align with the core objective: promoting safety and empowering users through intelligent navigation solutions.

- **Empathize:**
  - Conducting user surveys with users, especially women who travel frequently, to identify safety concerns with commute and travel.
  - Gathering insights from public surveys and platforms regarding the types of incidents that worry travelers and impact their travel decisions.

- **Define:**
  - The definition of a problem statement for GuardianNav is as follows:

    > "Users, especially women who travel alone in urban environments, often face safety concerns while navigating, leading to anxiety and avoidance of certain routes. Existing navigation apps focus on speed and distance but fail to consider real-time safety data and the ability to personalize routes for safety."

- **Ideate:**
  - Exploring ideas for incorporating crime data, feedback loops, and personalized safety metrics into the navigation system.
  - Exploring creative solutions for the user interface, ensuring cluster markings, safety scores, and route visualizations are easily accessible and intuitive.
- **Prototype:**
  - Building a basic version of the app's interface that highlights crime hotspots and includes visual indicators for route safety scores and safe vs. risky areas.
  - Developing the basic functionality of the app, i.e., providing safe navigation by integrating local crime data and personalized user preferences.
- **Test:**
  - Gathering feedback on the app's effectiveness of safety metrics, ease of use, design, and how well it addresses their safety concerns.
  - Iterating on the design based on user feedback and improving the performance of the algorithm with real-time updates and better data integration.

4.4 System Architecture

In the proposed architecture, GuardianNav is mainly comprised of a Flask-based front end for an intuitive interface, a Python-powered back end for data processing, and integration with external APIs such as the Open Route Service API for route data and DATA Police UK API for real-time crime statistics. By combining crime clustering, route optimization, and user preferences, the architecture provides personalized route recommendations. It balances safety and efficiency, empowering users with a secure and interactive navigation experience tailored to their needs.

Here is the detailed breakdown of every component of the System Architecture of Guardian-Nav:

4.4.1 Front-End: Flask Application

- **Framework:** Flask
- **Functionality:**
  - Provides a simple interface for users to input their journey start and endpoints.
  - Displays map visualizations with highlighted routes and safety information.
  - Allows users to view basic safety metrics, such as safety scores for routes.
  - Provides interactive route suggestions based on processed safety data.

The application constructed using the Flask framework enables the seamless rendering of the app and facilitates the creation of user-friendly web pages where users can input their journey details, such as start and end locations. It ensures smooth communication between the user and the back-end services, dynamically fetching and displaying route options alongside safety metrics. The simplicity and modularity that the framework provides help in the iterative development of the GuardianNav app.

4.4.2 Back-End Layer: Python-Based

- **Language:** Python
- **Responsibilities:**
  - **Data Fetching:** Integrate with:
    * Open Route Service API: To fetch navigation data and suggest potential routes.
    * DATA Police UK API: To gather real-time crime data based on geolocation.
  - **Data Processing:**
    * Analyse crime data for identified routes.
    * Calculate safety scores based on crime density, type of incidents, and user preferences.
    * Clean, manage, and optimize the dataset for computation and operations.
  - **Clustering and Optimization:**
    * Use clustering algorithm to group high-crime areas for better visualization and avoidance. K-Means algorithm is utilized for the effective clustering of crime hotspots.
    * Optimize route selection based on a balance of safety metrics and travel time.
  - **Dynamic Re-Routing:** Adjust routes in real-time if new crime data or incidents are reported.

- **User Preferences:** Process user inputs to customize navigation suggestions (e.g., prioritizing lit areas or avoiding specific neighborhoods).

The back-end layer, constructed using Python, is the core of GuardianNav and leverages Python's robust ecosystem to facilitate effective integration of machine learning and data processing. This layer is responsible for handling user inputs, fetching route data from the Open Route Service API, and retrieving crime statistics from the DATA Police UK API. Python's extensive libraries, such as NumPy and pandas, streamline data analysis, while frameworks like scikit-learn enable the development and deployment of machine learning models for safety metric calculations.

4.4.3 APIs Integration

- **Open Route Service API:**
  - Provides route data, including travel paths, waypoints, and distance/time estimates.
- **DATA Police UK API:**
  - Supplies crime statistics with geolocation data to identify risky areas along routes.

4.4.4 Data Processing

- **Framework:** Python
- **Current Processing Capabilities:**
  - Combines route data with crime data to calculate safety scores for different routes.
  - Maps crime incidents onto the route to highlight high-risk segments.
- **Processing Logic:**
  - **K-Means Clustering:** K-Means is an unsupervised machine learning algorithm used to group data points into clusters based on their similarity. It works by dividing data into K clusters, where each cluster is represented by its centroid. The algorithm iteratively minimizes the distance between data points and their respective centroids, ensuring that points within the same cluster are as similar as possible.
  - In GuardianNav, K-Means is applied to analyze and group crime data spatially. By clustering crime incidents based on their geographic coordinates, the algorithm identifies high-risk zones or "hotspots." These clusters allow the app to provide a clearer understanding of unsafe areas along potential routes. When evaluating

routes, GuardianNav can prioritize paths that avoid these high-crime clusters, enhancing user safety. Additionally, clustering helps optimize data visualization by highlighting risk areas, enabling users to make more informed navigation decisions.

4.4.5 Route Optimization

- **Current Logic:**
  - After identifying high-crime clusters along potential routes, the optimization process evaluates and ranks these routes based on their safety scores, which consider factors like crime density, type, and proximity to the route.
  - Using data from the Open Route Service API, the app combines travel efficiency (distance and time) with safety parameters to recommend the most secure and practical route. The optimization ensures a balance between minimizing travel risks and maintaining reasonable journey times, providing users with routes that prioritize their safety without significant inconvenience.

4.4.6 Output Layer

- Displays the safest route and alternatives with basic safety metrics. Highlights high-crime zones along the selected route for user awareness. Shows routes dynamically based on the processed data from APIs.

- In the context of GuardianNav, `Folium` is used to render the map on the front end, displaying the user's route, crime clusters, and safety information in an interactive format. `Folium` is a Python library used for creating interactive maps, built on top of the Leaflet.js JavaScript library. It allows easy integration of geographic data and visualizations within a web-based interface. `Folium` supports various types of map markers, popups, choropleth maps, and overlays, making it ideal for visualizing geospatial information, such as routes, locations, and crime hotspots.

- The functionality offered is as follows:

  1. **Route Visualization:** Displays the recommended route on the map, with clear markers for the start, end, and waypoints.
  2. **Crime Hotspot Highlighting:** Uses color-coded markers or polygons to highlight high-crime zones along the route, helping users avoid dangerous areas.
  3. **Safety Scores:** Shows safety metrics or scores for each route, possibly using tooltips or popups, giving users a clear idea of the risks associated with different paths.

4. **User Interaction:** Allows users to zoom, pan, and explore alternative routes or hotspots directly on the map for better decision-making.

4.4.7 Flowchart

The flowchart (Fig. 4.1) below shows the step-by-step process in determining the safest route by the application. It begins with collecting user inputs (start and destination coordinates) and processing crime data to identify high-risk zones using K-Means clustering. Routes are generated using the OpenRouteService API, and each waypoint is assessed for proximity to high-risk clusters. Unsafe routes are modified or recalculated to minimize overlap with risky areas. The safest route is then visualized on a map, along with high-risk zones and crime hotspots. Finally, risk-related statistics are displayed, and the results are saved as an interactive HTML file for user interaction and analysis.

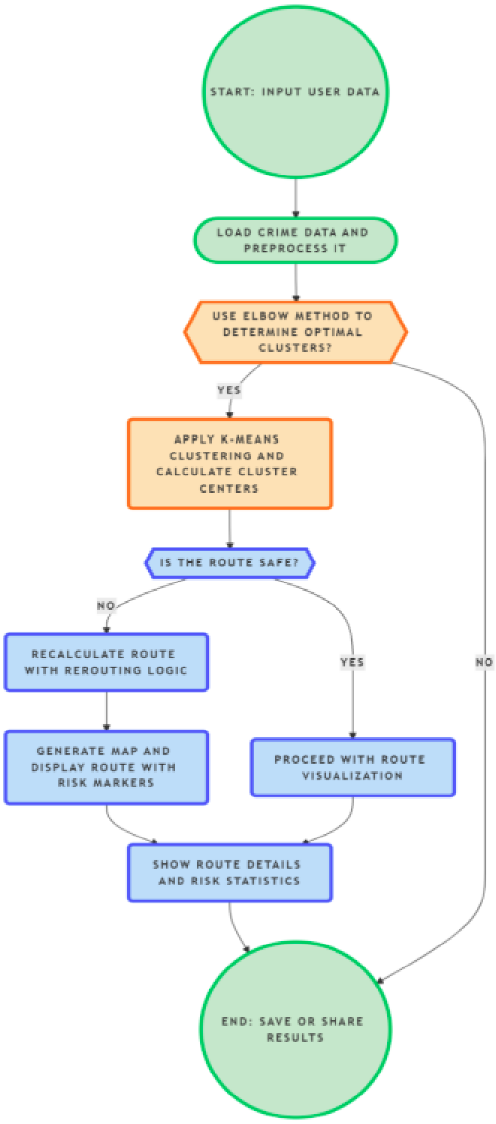

Figure 4.1: Flowchart of Application

4.4.8 Algorithm of the Application

This algorithm displays a detailed representation of the workflow of the application GuardianNav.

Input Variables:

- **Start Location (S):** User-provided starting point.
- **End Location (E):** User-provided destination.
- **Crime Data (C):** Data fetched from DATA Police UK API, containing crime type, coordinates, and timestamp.
- **Route Data (R):** Routes between S and E fetched from Open Route Service API.
- **User Preferences (P):** Optional user-defined safety preferences (e.g., avoid certain areas).

4.5 Expected Output

- **Safe Route**: A recommended route with a calculated safety score and highlighted risky areas.

4.6 Algorithm

- **Step I: Initialization of User Input**: Fetching the source and destination coordinates from the user to return the safe route between them.
- **Step II: Fetching of Route Coordinates**: Use the Open Route Service API to get possible routes $R = \{R_1, R_2, \ldots, R_n\}$ between source S and destination E. Each route R_i is represented as a series of waypoints $W = \{W_1, W_2, \ldots, W_m\}$.
- **Step III: Fetching Crime Data**: The crime data is fetched from the UK Police API based on the area of jurisdiction that the source and destination fall under.
- **Step IV: Calculation of Safety Score of Routes**: The safety score for each route in GuardianNav is calculated by analyzing crime data along the route, considering factors like crime density, type, and proximity to the route. Each crime event is assigned a weight based on severity, with more serious crimes (e.g., assault) impacting the score more significantly.

$$S_i = \frac{1}{(\text{Total Crimes})} \sum_{k=1}^{n} w_k \cdot T_k$$

- **Step V: Optimize Route Selection**:
  - **Safety Score** S_i: Higher scores preferred.
  - Select the safest route based on safety.
- **Step VI: Highlight Risky Areas**:
  - Risky areas and crime hotspots are marked based on the K-Means Clustering algorithm performed on the crime data fetched from local crime data.
  - These areas are programmed to be avoided by modifying the safety scores based on proximity to these areas.
- **Step VII: Display Safe Path**:
  - The safest route is displayed based on the above-discussed operations and functions.
  - The output map rendered includes the safe route and the areas that are determined crime hotspots and unsafe zones.

4.6.1 Data Processing

Numerous categories of data need to be processed and handled during the working of the application. According to the principles of data mining, there are stages that the data from various sources must undergo to promote ease of operability and uniformity.

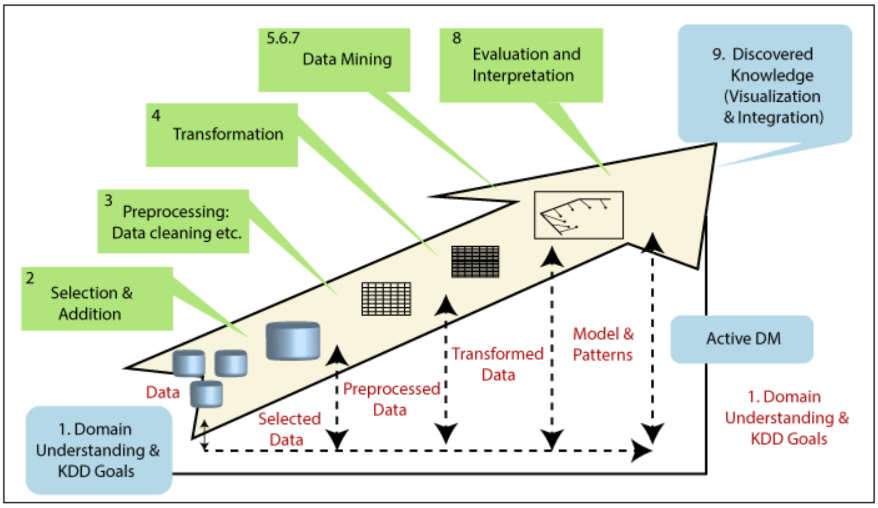

Figure 4.2: Data mining process

Fig 4.2 shows the Knowledge Discovery in Databases(KDD) process to derive meaningful patterns from datasets.

1. Data Selection:

- **Crime Data:** The first step involves selecting relevant crime data from sources like the DATA Police UK API, user reports, and historical crime databases.
- **Route Data:** Data from the Open Route Service API is also selected to generate various route options.
- **User Data:** User preferences, such as avoiding specific areas or preferences for route speed vs. safety, are gathered.

2. Data Preprocessing:

- **Cleaning:** This step ensures that the data is clean, removing any duplicates, inconsistencies, and missing values from crime data, route data, and user reports. Incomplete data, like missing coordinates or erroneous crime records, is filtered out.
- **Normalization:** Crime data is normalized (e.g., scaling crime counts or adjusting for different crime types) to ensure it can be compared across different areas and time periods.

- **Geotagging:** Crime data needs to be geotagged to match it with geographical coordinates, making it possible to visualize the locations of incidents on a map.

3. **Data Transformation:**
   - **Feature Extraction:** From the cleaned data, features such as crime density, severity, and proximity to route waypoints are extracted. These features are critical for route safety scoring.
   - **Clustering:** Using the K-Means algorithm, the data is grouped into clusters of high-crime areas (hotspots). This clustering allows the identification of regions with concentrated crime incidents, which are then used for safety analysis along routes.

4. **Pattern Evaluation:**
   - **Safety Score Calculation:** After applying mining techniques, the safety scores for each route are evaluated based on the detected patterns and clusters of crime data. The evaluation assesses which route is the safest based on the crime distribution and severity.
   - **User Preferences:** These patterns are refined further by integrating user-specific preferences, such as avoiding certain areas, to offer personalized route recommendations.

5. **Knowledge Representation:**
   - **Route Visualization:** The final mined knowledge is displayed to the user via interactive maps created using Folium. This visual representation shows the safest routes, highlighting high-crime zones, and provides safety scores for each path.
   - **Dynamic Updates:** Real-time crime data may lead to dynamic updates in route recommendations, allowing the app to continuously adjust and suggest the safest path based on new patterns in crime data.

4.7 Key Features of the Data Processing Pipeline

This pipeline integrates multiple techniques, including K-Means clustering for identifying high-crime areas and Machine Learning for personalized safety score prediction. The system continuously processes and updates crime data, user preferences, and environmental factors to generate dynamic and contextually relevant route recommendations. By leveraging these techniques, GuardianNav ensures that its users receive real-time, data-driven insights for safe navigation, tailored to individual needs and preferences.

4.7.1 K-Means Algorithm in GuardianNav

The K-Means clustering algorithm is a fundamental tool used to identify high-risk crime areas in the geographical data of GuardianNav. K-Means is an unsupervised learning algorithm that groups data points into clusters based on similarity. In this case, crime incidents are grouped based on geographical proximity.

How K-Means Algorithm Works in GuardianNav:

1. **Crime Data Clustering:** The K-Means algorithm processes crime data (latitude and longitude coordinates of crime incidents) to form clusters of high-crime areas (hotspots).

    - Initially, the number of clusters (k) is chosen based on the level of granularity needed (for example, identifying neighborhoods or smaller areas with concentrated criminal activity).
    - The algorithm then groups the crime data into k clusters, with each cluster representing a hotspot or a region with a high concentration of crimes.

2. **Clustering High-Risk Areas:** After the data is clustered, each cluster corresponds to an area with a relatively higher crime density. These clusters are critical for the app as they highlight the areas of concern along any given route.

3. **Route Impact:** Routes passing through these clusters (high-crime areas) are assigned lower safety scores. Routes that avoid these clusters or pass through low-crime areas are given higher safety scores.

4. **Adjusting Clusters Over Time:** As new crime data is collected or user reports are added, the K-Means algorithm can be rerun to update the clusters. This ensures that the clustering reflects the most current crime data, allowing GuardianNav to dynamically update the safest routes.

Integration of K-Means and Data Processing with GuardianNav

In GuardianNav, the integration of K-Means clustering and data processing plays a crucial role in improving the accuracy and effectiveness of safety recommendations, specifically for routing in high-risk areas. Here's a breakdown of how K-Means and the data processing flow work together to ensure safe navigation:

1. **Crime Data Processing with K-Means Clustering:**

    - K-Means clustering is used to process crime data by grouping incidents based on their geographical proximity. This clustering technique identifies areas with high concentrations of crime, which are flagged as high-risk zones.

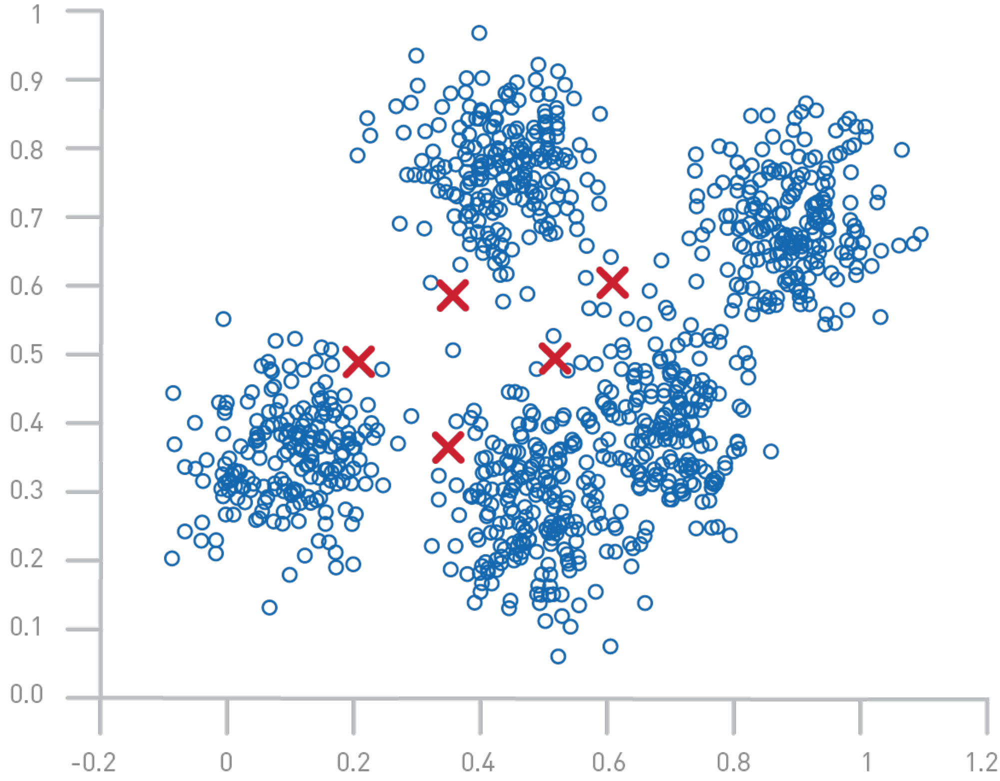

Figure 4.3: Example of K-Means Clustering

- The clusters formed from this analysis provide a clear view of crime hotspots in the city, making it easier for the system to pinpoint regions with elevated risks. These clusters become the foundation for the safety recommendations and route scoring in the application.

2. **Integration with Route Data for Safety Optimization:**

   - The clusters generated by K-Means are integrated with route data to optimize navigation. The system evaluates potential paths and recommends safer alternatives by avoiding high-risk clusters identified through the crime data.
   - Real-time updates are crucial for maintaining accuracy in route suggestions. As new crime data is incorporated into the system, the clusters are recalculated, allowing GuardianNav to adjust route recommendations dynamically, ensuring that users are always provided with the safest options available.

3. **Data Processing Flow Without Machine Learning Models:**

   - Unlike approaches that involve machine learning models for prediction or learning from user data, GuardianNav relies on K-Means clustering for categorizing crime data. The output from clustering serves as a direct input for generating safe routes and crime risk evaluations.

- The system focuses on periodic updates to crime clusters rather than continuous model training, making it more efficient in processing real-time crime data without the need for complex machine learning algorithms.

By utilizing K-Means clustering for categorizing crime incidents and optimizing routes, GuardianNav ensures safety while maintaining simplicity and efficiency in its data processing pipeline.

4.8 Final Solution

- **Polished App:** GuardianNav delivers a robust navigation platform that combines precise safety analysis with seamless route planning, providing users with a reliable and user-friendly experience.
- **User Testing:** Feedback from initial users highlights significant safety improvements, with users particularly appreciating the app's focus on avoiding high-risk zones and its user-centric design.
- **Scalability:** The system is built to handle large-scale expansion, allowing integration of additional datasets and enabling deployment in other cities, adapting to local safety requirements effortlessly.

4.9 Impact and Feasibility

4.9.1 Impact:

- GuardianNav empowers women by increasing their confidence and mobility, making urban navigation safer and more inclusive.

4.9.2 Social Benefit:

- Encourages safer urban environments through community-driven reporting and real-time alerts, fostering collective responsibility for public safety.

4.9.3 Feasibility:

- **Adoption Potential:** Designed with urban populations in mind, ensuring high usability and demand in densely populated areas.

4.10 Challenges and Lessons Learned

4.10.1 Challenges:

- Variability and inconsistency in crime data and real-time data.
- Balancing the dual priorities of safety and travel efficiency in route recommendations.

4.10.2 Lessons Learned:

- Real-time data integration is crucial for maintaining the accuracy and relevance of navigation systems.
- Regular user feedback and iterative development cycles are essential for refining the app and improving user satisfaction.

Chapter 5

IMPLEMENTATION

This document outlines the implementation of GuardianNav: ML-Driven Safe and Personalized Navigation for Women designed to enhance user safety, particularly for women, by leveraging a combination of Flask, machine learning, geospatial clustering, and real-time crime data analysis. The core objective of GuardianNav is to provide users with personalized, safer route recommendations by integrating multiple data sources, including real-time crime records, traffic data, and historical safety information.

5.1 Experimentation

GuardianNav was implemented using a combination of Flask, machine learning, geospatial clustering, and real-time crime data analysis. Below is an overview of the steps involved in the implementation process:

1. **Data Collection and Preprocessing:** The system loads crime data from multiple CSV files stored in a specified directory. These files include crime records with information such as the type of crime and its location (latitude and longitude). After loading the data, it is filtered to include only relevant crime types, such as violence and sexual offenses. Any rows with missing latitude or longitude values are dropped to ensure accurate geospatial analysis.

2. **Clustering High-Risk Zones:** KMeans clustering is applied to the filtered crime data to identify high-risk zones. By analyzing the geographic coordinates of crimes, the algorithm groups the data into clusters. The system then calculates the centroids of these clusters, which represent high-risk areas that should be avoided during route generation.

3. **Route Generation:** The system utilizes OpenRouteService (ORS) to generate routes between user-specified start and end points. For every route generated, the system checks each waypoint to ensure it is not too close to any high-risk cluster. If any waypoints fall within the vicinity of a high-risk area, those points are excluded, and a safer alternative route is calculated.

4. **Geolocation and Address Resolution:** Geopy's Nominatim API is a powerful tool used to convert user-inputted addresses into accurate geographic coordinates (latitude and longitude). This geocoding service is essential for GuardianNav, as it allows the

system to effectively map user-provided locations to precise points on the globe. The ability to translate addresses into coordinates ensures that the system can dynamically generate accurate routes between a user's start and end locations. Without this functionality, the system would not be able to interact with OpenRouteService (ORS) to calculate the optimal paths.

5. **Real-Time Map Rendering:** Folium is used to render the generated routes and crime locations on an interactive map. The map displays:
   - The safe route in green.
   - Start and end locations with markers.
   - High-risk areas represented as red circle markers.
   - Individual crime locations displayed as orange circle markers.
6. **User Interface:** The frontend of the system is designed using HTML and JavaScript. Users can input their start and end addresses into a form. Upon submission, the form sends a POST request to the Flask backend, which processes the data and returns an interactive map with the safe route. The map is rendered dynamically using Folium and displayed within the webpage.

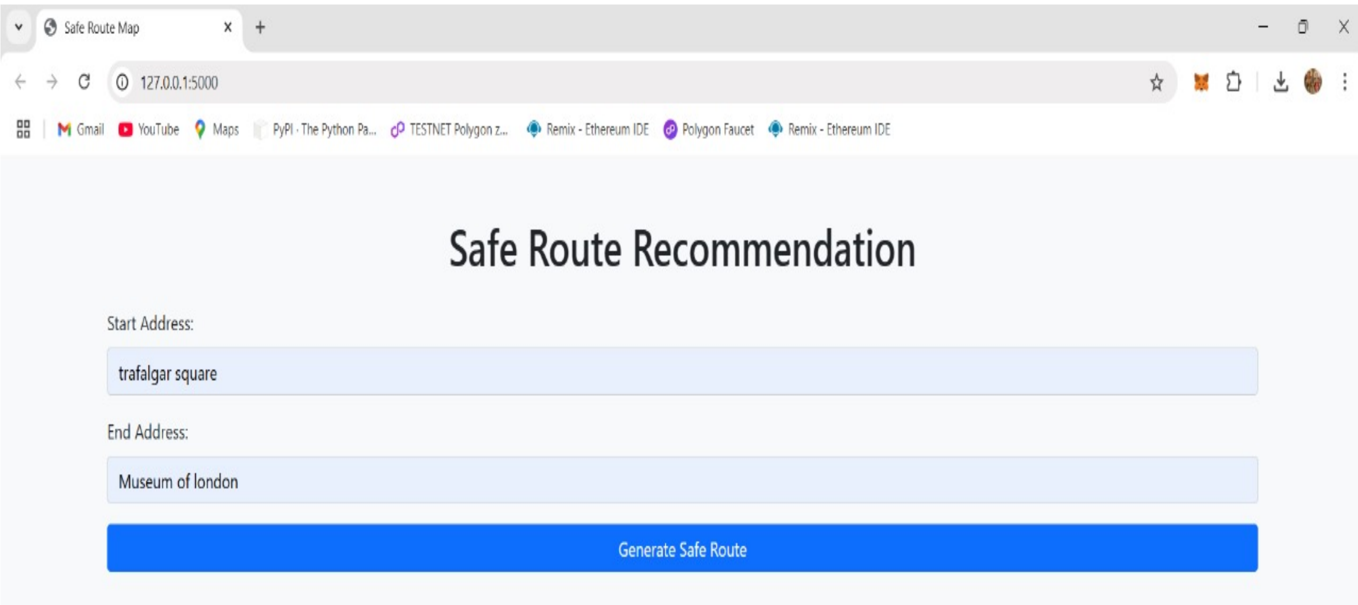

Figure 5.1: User Interface

Fig 5.1 shows the GuardianNav User Interface facilitating the entry of start and end locations to determine safe route.

5.2 User Interface

The GuardianNav user interface is designed to be intuitive and user-friendly. It consists of the following components:

1. **Address Input Form:** Users can input their start and end addresses into text fields. Behind the scenes, these addresses are geocoded to obtain latitude and longitude coordinates for further processing.

2. **Interactive Map:** After submitting the form, an interactive map is displayed showing the safest route, crime locations, and high-risk zones. The map is embedded in the webpage using Folium, providing users with a visual representation of their journey and safety along the route.

3. **Error Handling:** If the addresses are invalid or the route cannot be generated due to unavailable data, the user receives an appropriate error message, ensuring smooth interaction and clarity of functionality.

4. **Map Features:**

   - Safe route: The route is displayed in green, with dynamic updates showing safe path alternatives.
   - Crime areas: Crime locations are shown as orange markers, and high-risk areas are indicated by red circle markers.

5.3 Result and Analysis

The GuardianNav system provides real-time, personalized safety recommendations based on the analysis of historical crime data and machine learning techniques. Here's a breakdown of the results and analysis:

1. **Route Safety Evaluation:** The system successfully analyzes the user's route and avoids high-risk areas identified through clustering. For several test cases, the generated routes were validated against known crime data, and the system consistently suggested safer alternatives by avoiding high-risk zones. The analysis of crime data using clustering techniques (KMeans) allowed for accurate identification of high-crime zones, effectively preventing users from traveling through these areas.

2. **Accuracy and Performance:** The system demonstrated reliable performance when generating safe routes, with minor latency introduced by the geocoding and route generation processes. The KMeans clustering algorithm, while effective, can be further improved with more precise clustering parameters, depending on the density of crime

data. The real-time map rendering provided a seamless user experience, ensuring that route recommendations were clearly visualized.

5.4 Limitations

- **Data Dependence:** The accuracy of the recommendations depends heavily on the quality and completeness of crime data. In areas with sparse data, the clustering model may not be as effective.
- **Real-Time Data:** While the system integrates historical crime data, it does not yet account for real-time incidents or ongoing crimes, which could further improve route safety in dynamic environments.
- **Scalability:** As the system processes more data and traffic, the backend's performance could be impacted, particularly during high request volumes. Further optimization and cloud integration could help scale the system.

5.5 Future Work

Future versions of GuardianNav can incorporate real-time crime data integration from law enforcement APIs or social media, enhancing the accuracy of safety recommendations. Moreover, expanding the system to include different transportation modes (walking, cycling, etc.) could broaden the usability of the platform for various user needs. Additionally, user feedback mechanisms for crowd-sourcing incident reports could further refine the system's accuracy and real-time response capabilities.

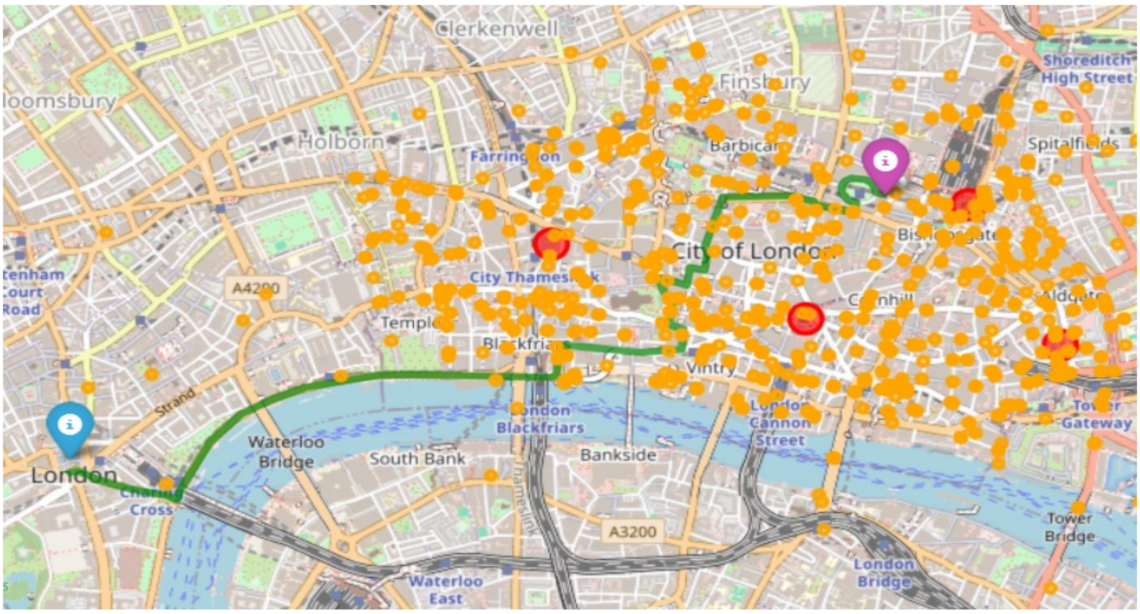

Figure 5.2: Result Obtained

Fig 5.2 displays the map representation of the safe route recommendation provided by GuardianNav to the prescribed input.

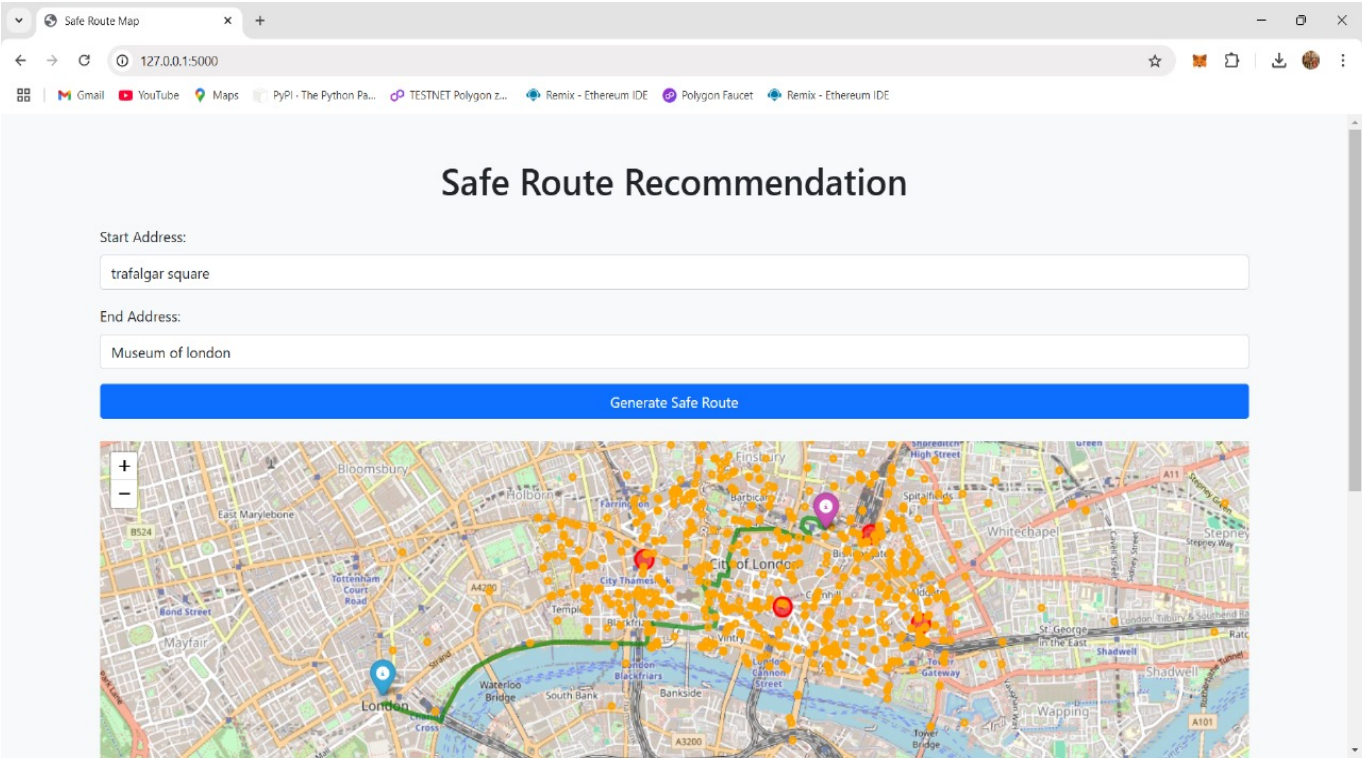

Figure 5.3: Complete UI Design

Fig 5.3 shows the complete User Interface in the users point of view showcasing clean and crisp input and output flows.

Chapter 6

Conclusion and Future Work

GuardianNav integrates machine learning, geospatial clustering, and real-time crime data to offer personalized and safer route recommendations, prioritizing user safety during urban travel. By leveraging KMeans clustering for identifying high-risk zones and using OpenRouteService for route generation, the system ensures that users can navigate through safer paths, avoiding areas with higher crime rates. The use of Geopy's Nominatim API allows precise geolocation and routing, making the system reliable for real-time navigation. Through its interactive maps, GuardianNav provides users with visual feedback on crime locations and high-risk zones, enhancing the overall user experience.

To enhance GuardianNav, future versions can incorporate real-time crime data from law enforcement APIs or social media, providing more accurate and timely safety recommendations. Expanding the system to include multiple transportation modes like walking and cycling would broaden its usability, catering to a wider user base. Additionally, integrating crowdsourced incident reports could further refine the system's real-time response capabilities, improving route recommendations based on user-generated data. These upgrades would make GuardianNav a more adaptive, dynamic, and comprehensive safety tool for urban travelers.

Chapter 7

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APPENDICES

Appendix A
SAMPLE CODE

```

import folium
from geopy.distance import geodesic
import openrouteservice
import pandas as pd
from sklearn.cluster import KMeans
import os

# OpenRouteService client setup
ORS_API_KEY = '########################################'
client = openrouteservice.Client(key=ORS_API_KEY)

# Data folder and crime data
data_folder = '/content/drive/MyDrive/dtset'
all_files = [os.path.join(data_folder, f) for f in os.listdir(
    data_folder) if f.endswith('.csv')]
crime_data = pd.concat((pd.read_csv(f) for f in all_files),
    ignore_index=True)

# Filter crime data for relevant crimes
filtered_data = crime_data[crime_data['Crime type'] == 'Violence and
     sexual offences']
filtered_data = filtered_data.dropna(subset=['Latitude', 'Longitude
    ']).reset_index(drop=True)

# KMeans clustering
coordinates = filtered_data[['Latitude', 'Longitude']]
kmeans = KMeans(n_clusters=5, random_state=0)
filtered_data['Cluster'] = kmeans.fit_predict(coordinates)
cluster_centers = kmeans.cluster_centers_

# Function to check proximity to clusters or crimes
def is_near_cluster(coord, clusters, crime_data, radius=0.5):
    for cluster in clusters:
```

```
        if geodesic((coord[1], coord[0]), cluster).km < radius:
            return True
    for _, crime in crime_data.iterrows():
        if geodesic((coord[1], coord[0]), (crime['Latitude'], crime['Longitude'])).km < radius:
            return True
    return False

# Function to calculate safe route
def get_safe_route(start_coords, end_coords, clusters, crime_data, radius=0.5):
    try:
        route = client.directions(
            coordinates=[start_coords, end_coords],
            profile='driving-car',
            format='geojson'
        )
        route_coords = route['features'][0]['geometry']['coordinates']

        # Filter route waypoints to avoid high-risk zones
        safe_route_coords = []
        for coord in route_coords:
            if not is_near_cluster(coord, clusters, crime_data, radius):
                safe_route_coords.append(coord)

        # If all waypoints are unsafe, fallback to the original route
        if not safe_route_coords:
            print("All waypoints are near high-risk zones. Returning the original route.")
            return route_coords

        return safe_route_coords
    except Exception as e:
        print("Error fetching route:", e)
        return []

# Start and end coordinates
start_coords = [-0.09, 51.515]  # [longitude, latitude]
```

```
end_coords = [-0.1278, 51.5074]

# Get the safest route
safe_route_coords = get_safe_route(start_coords, end_coords,
    cluster_centers, filtered_data, radius=0.5)
safe_route_coords_corrected = [[coord[1], coord[0]] for coord in
    safe_route_coords]  # Convert to [latitude, longitude]

# Map visualization
route_map = folium.Map(location=[start_coords[1], start_coords[0]],
    zoom_start=13)

# Add the safe route
if safe_route_coords_corrected:
    folium.PolyLine(safe_route_coords_corrected, color="green",
    weight=5, opacity=0.7, tooltip="Safe Route").add_to(route_map)
else:
    print("No safe route calculated.")

# Add start and end points
folium.Marker(location=[start_coords[1], start_coords[0]], icon=
    folium.Icon(color='blue'), tooltip="Start Location").add_to(
    route_map)
folium.Marker(location=[end_coords[1], end_coords[0]], icon=folium.
    Icon(color='purple'), tooltip="End Location").add_to(route_map)

# Add clusters
for cluster in cluster_centers:
    folium.CircleMarker(
        location=[cluster[0], cluster[1]],
        radius=10,
        color="red",
        fill=True,
        fill_opacity=0.6,
        tooltip="High-Risk Area"
    ).add_to(route_map)

# Add individual crime locations
for _, row in filtered_data.iterrows():
    folium.CircleMarker(
        location=[row['Latitude'], row['Longitude']],
```

```
        radius=3,
        color="orange",
        fill=True,
        fill_opacity=0.5,
        tooltip="Crime Location"
    ).add_to(route_map)

# Save the map to an HTML file
route_map.save("safe_route_map.html")
route_map
```

Appendix B
CLIENT SATISFACTION REPORT

VELAGAPUDI RAMAKRISHNA SIDDHARTHA ENGINEERING COLLEGE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

EPICS CLIENT SATISFACTION REPORT

The project *"GuardianNav: ML-Driven Safe and Personalized Navigation"* focuses on providing safe and efficient route recommendations by leveraging advanced machine learning techniques, crime data analysis, user reports, and real-time incident updates. Developed by the students of Velagapudi Ramakrishna Siddhartha Engineering College, Vijayawada, the application is designed to enhance the safety and convenience of its users, with a particular emphasis on addressing the needs of women navigating unfamiliar areas.

For a naive user, the app provides a straightforward interface to input their starting point and destination. Based on this, it intelligently suggests the safest routes by analyzing crime data, user reports, and real-time incidents. The app highlights areas to avoid and ensures that users feel confident and secure while navigating, making it an essential tool for daily commutes and exploring new areas.

Guide Details:

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Peetha Manikantha (238W5A0515)

22/11/24

Client Signature

SUB - INSPECTOR
NANDIGAMA PS.

Figure 7.1: Client satisfaction report